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Senior Android Engineer · *Ex-Uber (contract)*
Android TV, Fire OS and multi-brand OTT streaming — Media3/ExoPlayer, DRM, and native JNI/C++ (Uber Maps SDK).

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§ 01 Summary

PROFILE · POSITIONING

Senior Android engineer with cross-stack reach — Uber Maps SDK (JNI/C++), OTT / CTV streaming frameworks, broadcast-grade DRM. Ten-plus years shipping production apps across Android, Android TV, Fire OS, webOS and Tizen — including a multi-brand OTT streaming framework and 4K DRM-protected playback (Widevine, PlayReady).

§ 02 Experience

ENGAGEMENTS · 2014 → PRESENT

Nexstar Media Group, Inc.

Aug 2025 — Present · current

Senior Fire OS / Android TV Engineer · via Nimble.LA · full remote CONTRACT

- Contributed to the architecture of an enterprise **OTT streaming framework** powering **multiple TV station brands** across Fire TV and Android TV using **MVVM + Clean Architecture** with a plugin-based integration layer for third-party services.
- Built five specialised **Media3 / ExoPlayer** implementations covering VOD, live, and audio playback with HLS adaptive bitrate.
- Shipped DRM-protected playback (**Widevine, PlayReady**) with both server-side and client-side ad insertion.
- Integrated the platform's analytics and audience-measurement stack under **GDPR / CCPA** compliance.
- Delivered TV-optimised auth (device-based), entitlement & flexible paywall, and zero-downtime feature rollouts across a large multi-brand flavor matrix via remote config; built Leanback TV UI with full d-pad navigation.
- Refactored callback chains to reactive Kotlin **SharedFlow**; resolved fragment-lifecycle leaks; wired **Firebase Crashlytics + New Relic** with JUnit, Mockk, Robolectric, Espresso coverage.
- Run an **agent-driven delivery workflow** end to end: an agent picks up the tracker issue with its acceptance criteria, works the change on an isolated branch, runs the test suite alongside agent-driven end-to-end checks against a running build, and opens a **GitHub** pull request carrying the issue context and test evidence. **GitHub Actions** runs the pipeline; a second model audits the diff against the original request before anything is reported done.

`Kotlin` `Coroutines / Flow` `Media3 / ExoPlayer` `Dagger 2` `Retrofit2` `Moshi` `Glide` `Android Leanback` `Jetpack (Nav, ViewModel, Room)` `Firebase` `Google IMA` `New Relic` `JUnit · Mockk · Espresso`

FreeCast, Inc.

Dec 2024 — Jul 2025 · 8 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Returned to the Select TV and FreeCast Watch products across Android, Android TV, Fire OS, webOS and Tizen through to the end of the engagement.
- Used **coding agents as a primary development method** on migration and refactor work, with automated verification in CI before review.

Claro TV+

Apr 2024 — Dec 2024 · 9 mo

Android TV / Fire OS Engineer · full remote CONTRACT VIA GLOBAL HITSS

- Greenfield Android TV / Fire OS app for one of Brazil's largest pay-TV operators; **4K streaming** via DASH / HLS with multi-DRM.
- Contributed to the data layer as a shared **Kotlin Multiplatform (KMP)** submodule consumed by Android, webOS and Tizen clients.
- Multi-module **MVVM** architecture; Media3 playback; analytics across Firebase + Google Analytics 4.
- Applied **AI-assisted development** across the greenfield build — code generation and review over the Kotlin Multiplatform and multi-module layers.

`Kotlin Multiplatform (KMP)` `Kotlin` `Jetpack Navigation · ViewModel · Room` `Hilt` `Media3` `Coroutines / Flow` `Firebase + GA4` `webOS · Tizen`

FreeCast, Inc.

May 2022 — Apr 2024 · 1 yr 11 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Led the **full migration of the legacy Java codebase to Kotlin**; introduced Use-Case + Clean Architecture and improved end-user UX.
- Implemented **HLS / DASH** playback for **live linear TV and VOD** with Widevine DRM-protected streaming; integrated Google Cast and a paid subscription gateway.
- Maintained and developed companion apps on **webOS and Tizen** alongside the Android TV / Fire OS surfaces.
- Integrated **OTA tuner platforms** (HDHomeRun, AirTV, Alticast, PDAQ) with GStreamer-based media pipelines for secure live broadcast over HLS / DASH.
- Owned **DRM, EPG, tuner orchestration, transcoding**, and multi-device streaming end-to-end.
- Shipped on Google Play (*Select TV Android, FreeCast Watch*) and Amazon (*FreeCast Watch Fire OS*).
- Used **AI-assisted development** on the Java-to-Kotlin migration and on widening test coverage.

`Kotlin` `Jetpack Compose` `Hilt` `Coroutines / Flow` `Media3` `Room · DataStore` `Retrofit2` `Picasso` `Firebase Analytics · Crashlytics · FCM` `Mixpanel` `webOS · Tizen` `HDHomeRun · AirTV · Alticast · PDAQ` `GStreamer` `Widevine DRM`

Uber Technologies

May 2021 — May 2022 · 1 yr

Android Engineer · Maps SDK team · via Matchpoint Solutions · full remote CONTRACT

- Worked on the **Maps team SDK** module — JNI/C++ bindings between Uber's native Maps engine and the Android client.
- Contributed to Uber's open-source Android architecture framework (**RIBs**); shipped to Uber Rider, Uber Eats and Uber Driver.

- Operated inside a 10K+ engineer codebase using **Buck, Flipper**, custom analytics / logger / crash reporter and a feature-by-module architecture.

`Java · Kotlin` `C++ · NDK · JNI` `Groovy` `Python · SQL` `RIBs` `Buck` `CMake` `Flipper` `Firebase`

FreeCast, Inc.

Mar 2021 — May 2021 · 3 mo

Android / Android TV / Fire OS Engineer · full remote CONTRACT

- Opening engagement on the Select TV Android and FreeCast Watch codebases, starting the migration from Java to Kotlin.

Super Revendedores

Oct 2018 — Feb 2021 · 2 yrs 5 mo

Android Engineer · full remote

- Migrated the full codebase from Java to Kotlin; implemented **online payment gateway**, virtual showcases and an SSO link to a partner web ordering portal.
- Built scalable asynchronous order processing on **AWS SQS / SNS** for reliable queuing and notification delivery.

`Kotlin` `AWS (SQS · SNS · Storage)` `Firebase (Firestore · Auth · Remote Config · IAM · FCM)` `Facebook OAuth` `DbFlow` `Dagger 2` `MVP` `Retrofit2`

Concrete · an Accenture Company

Dec 2016 — Oct 2018 · 1 yr 11 mo

Jr. Android Engineer

- Built the **Minha CVC** travel app from its initial implementation as part of the agency team — **1M+ downloads**; shop geolocation directing customers to physical stores; itinerary feature reducing customer-service call volume.
- Worked on **Getnet**; migrated MVP to MVVM; introduced RxJava2 and LiveData.

`Java 8` `Kotlin` `RxJava2 · LiveData` `MVP → MVVM` `Dagger 2` `Google Maps API` `Retrofit2 · Picasso` `Firebase` `Jenkins CI/CD`

PraPagar (fintech)

Jun 2015 — Dec 2016 · 1 yr 7 mo

Jr. Android Engineer · collaborating with US team (MaxxPotential)

- Built mobile payments via **QR code and geolocation**; first Android + Laravel REST experience.

`Android` `PHP · Laravel 5` `REST` `OOP · Design Patterns`

IBM Brasil

Jun 2014 — Dec 2014 · 7 mo

Software Engineering Intern · Rio de Janeiro · while pursuing B.S. at Instituto Infnet

- Wrote **SQL queries against IBM DB2** in support of contract and sales teams; produced reporting spreadsheets and Hyperion-based BI artefacts.
- Documented internal processes via SharePoint; first professional exposure to enterprise databases, REST APIs, and Agile workflows.

`IBM DB2` `SQL` `IBM Hyperion` `SharePoint` `Python` `REST APIs` `Git`

FASOLTI · Fábrica de Soluções em TI

Jan 2014 — May 2014 · 5 mo

Test Developer · INTERNSHIP

- Documented and implemented automated test cases in **Selenium with TestNG**.

AI-Native Engineering Practice

OWN FLEET INFRASTRUCTURE · FIGURES READ FROM SOURCE, NOT READMES

claude-dispatch — multi-provider LLM orchestration mesh

A **14,168-line** Python dispatch daemon routing AI coding work across a seven-machine fleet, exposed to the orchestrating agent as **15 MCP tools**.

- Cost-aware routing across **8 providers, 22 models, 6 tiers**: one resolver applies rate-limit backoff, per-provider cooldowns, parallel-safety and required capabilities in order. Task type is inferred from the prompt; a complexity score forces a request up a tier or down to a free one.
- Automatic escalation deliberately stops below the expensive tier — the orchestrator is never dispatched as a worker, and frontier tiers are unreachable from any automatic path.
- Flat-rate providers are marked **serial-only**, so a parallel fan-out physically cannot select one and burn its concurrency limit.
- Encrypted job transport keyed off the existing SSH identity — **RSA-OAEP + AES-256-GCM**, fresh key and nonce per message, clients authenticated by public-key fingerprint. No second PKI to operate.
- **Five-source peer discovery** (mDNS/Bonjour, config, Tailscale, peer exchange, gossip heartbeats) under an explicit precedence ladder; per-job token and cost extraction with live WebSocket output.

Encrypted clipboard & screenshot mesh

52,097 lines, 1,624 tests. Hybrid AES-256-GCM + RSA-OAEP over the same SSH identity.

- **Four separate HMAC token families** — clip, API, share, archive — scoped so a share token cannot be replayed to fetch a different artefact.
- Peer-to-peer by default with a per-machine store; a seven-method access cascade falls from local HTTP through LAN, SMB, SCP, SFTP and Tailscale before any cloud path. Decryption is server-side: a recipient needs the token, never the key.

Manifest-driven fleet deployment

A supervisor running unattended across **five machines** under launchd, systemd and Task Scheduler, managing **eleven applications**.

- Each app ships a monotonic `VERSION.json` and a manifest declaring its install, launch and `/health` contract — deployment is a git push.
- Supervision and auto-update run on deliberately decoupled intervals, so a crash restarts in seconds without waiting for the update poll. Cross-machine control is RSA-signed per request; failed self-updates roll back and queue a pending action.

Instrumentation as a definition of done

I authored a language-neutral **trace-contract** standard: the compile-time reference graph is the specification, and the runtime trace is reconciled against it automatically.

- Ships a written spec, JSON Schemas, a cross-language conformance oracle, a golden-case corpus, and an agent plugin whose hook blocks completion while the graph is stale. Three implementations across Python and TypeScript services, generating live trace volume in production.
- Gate severity is scoped rather than uniform: agent-authored edits are blocked at the definition of done, while human commits degrade to a lint error by profile — never a unit-test failure.

Closed-loop evaluation on real devices

For TV clients I expose an on-device HTTP diagnostic surface — trace, state, screenshot, scripted navigation, and a named test-routine harness of **149 routines**.

- Each routine declares required parameters and returns a structured pass/fail with per-step detail, or HTTP 422 with a machine-readable reason. An agent drives the app, captures what the screen actually shows, reads the emitted events and evaluates against the contract — no human in the loop.
- The same discipline applies to the harness: routines snapshot and restore application state on every exit path, and health probes are excluded from idle accounting — a monitoring call must not be mistaken for activity by the thing it monitors.

Resource governance

A GPU inference front door that verifies a model is fully resident after load, unloads it and fails with a typed error rather than allowing a silent spill to CPU — measured **11.6 GB to 1.8 GB** reclaimed.

- Long-running jobs publish liveness-stamped progress files, where stalled is judged by heartbeat age rather than process existence — a wedged process keeps a live PID.

§ 04

Published Applications

APP STORE · GOOGLE PLAY · SHIPPED UNDER MY OWN DEVELOPER ACCOUNTS

Archive — Home Video Player

APPLE TV (TVOS)

Native SwiftUI tvOS client for a self-hosted media library: HLS playback, server-side transcoding, library browsing and resume across devices.

`Swift, SwiftUI, tvOS, AVFoundation, HLS`

Beer Price Calculator

ANDROID — GOOGLE PLAY

Compares beer prices by volume to find the best value per millilitre. Offline-first with local persistence and in-app billing.

`Kotlin, Jetpack Compose, Room, Play Billing`

Beer Price Calculator

IPHONE & IPAD (IOS)

Native SwiftUI build of the same price-per-volume comparison app for iPhone and iPad.

`Swift, SwiftUI, iOS`

§ 05

Selected Personal Work

INDEPENDENT · OPEN SOURCE · RESEARCH

Arquive — backend for the Archive tvOS app above

Self-hosted personal media archive & streaming server. In-browser **HLS playback** with GPU-accelerated transcoding cache, **IPTV** (M3U + XMLTV EPG +

DynamicMusicApp

Android music player with **three pluggable design systems** users switch at runtime — *Material You* (Google), *Neon Wave* (synthwave), and *Organic Flow* recording), DLNA/UPnP for smart TVs, distributed GPU-fleet transcoding, FAISS-backed face recognition, local-AI captions via Ollama, plus a native **Apple TV (SwiftUI)** client.

Python · React · HLS · GPU Transcoding · FAISS · InsightFace · Ollama · Qwen2.5-VL · DLNA · M3U · XMLTV · SwiftUI tvOS

(botanical). One Compose codebase, three radically different visual identities — design-system theming treated as a first-class architectural concern.

Kotlin · Jetpack Compose · Material 3 · MVI · Coroutines / Flow · Hilt

§ 06 Education

FORMAL · CONTINUING

Instituto Infnet · Rio de Janeiro — 2013 – 2018
B.S. Computer Engineering — software development, mathematical modelling, OS & hardware architecture.

Chiswick House School (Malta) — 1997 – 1998
Primary education (English-medium).

Orion Consulting · Rio de Janeiro — 2010
Java fundamentals — OOP, software design, code patterns and logic.

§ 07 Technical Stack

CORE COMPETENCIES

CORE	Android · Applied AI / LLM · Kotlin
ANDROID · TV	Android SDK · Android TV · Fire OS · webOS · Tizen · Jetpack Compose · Leanback · Kotlin Multiplatform · NDK · JNI
STREAMING	Media3 / ExoPlayer · HLS · DASH · Live · VOD · Widevine · PlayReady · SSAI · CSAI · Google IMA · GStreamer
ARCHITECTURE	MVVM · MVI · Clean Architecture · Use Case · Multi-module · Coroutines · Flow · Hilt · Dagger 2
AI SYSTEMS	Agent orchestration · MCP · RAG · vector search · Evaluation harnesses · Local inference · Python
DELIVERY	JUnit · Mockk · Robolectric · Espresso · Gradle · GitHub Actions · Jenkins · Jira